

Alahe Akhavan

 akhavana |  alaheakhavan |  Portfolio |  alahe.akhavan@gmail.com |  (279) 758-1676

EDUCATION

Johns Hopkins University

Master of Science in Mechanical Engineering

Baltimore, MD

August 2026 – May 2028

University of California, Berkeley

Bachelor of Science in Mechanical Engineering

Berkeley, CA

July 2023 – December 2025

EXPERIENCE

Assistive Grasping, Robotic Research Assistant

Embodied Dexterity Group (EDG), Advised by Prof. Hannah Stuart

May 2024 – June 2025

Berkeley, CA

- Worked with a group of graduate students as the **lead designer** in creating the next-generation assistive device, Dorsal Grasper, intended for people with C6/C7 SCI to enabling functional object manipulation in human trials.
- Integrated **electromechanical system** including custom PCB, Flex sensor, switches, and 3D-printed structure designed with SolidWorks CAD software for the Bambu printer for rapid prototyping.
- Improved structural stability through iterative testing of 3D-printed polymer materials, evaluated rigidity and strength to inform material and component selection.
- Contributed in **human-subject testing** across participants with varying SCI levels, research paper is currently under peer review. Presented poster presentations at SACNAS 2024 and ASME-IMECE 2024 Conferences.

Prosthetic Hand for Grasping and Manipulation, Research Fellow

Assistive Robotics and Manipulation Lab (ARMLab), Advised by Prof. Monroe

June 2025 – August 2025

Kennedy III

Stanford, CA

- Developed a **closed-loop control framework** for dexterous robotic manipulation of everyday objects using **multi-modal sensing** of bio-signal input and perception for the Inspire Robot Hand.
- Implemented real-time control strategies and **reinforcement learning** using Rokoko GLoVES for object manipulation, which improved robustness in articulated tasks.
- Created a multi-component 3D handle using SolidWorks CAD software for the Inspire Hand for future participant testing, performed FEA analysis to ensure robustness.

Tremor Glove (Biomedical Device), Undergrad Research

Advised by Prof. Grace O'Connell

August 2024 – July 2025

Berkeley, CA

- Collaborated with a team of engineers to design a wearable assistive glove to suppress tremors using mechanical stabilization method of varying weights across the hand.
- Utilized co-design method and worked with a participant from the start, took feedback and created a device fit for the users. Used **machining and precision engineering** to design the glove.
- Tested our design with participants for daily activities, comparing with several devices on the market. Our tremor glove allows for a more smooth transition of grasp-to-manipulate with users.

Tactile Sensing Control Optimization, Research Assistant

Embodied Dexterity Group (EDG), Advised by Prof. Hannah Stuart

January 2024 – April 2024

Berkeley, CA

- Optimized the **UR10 Robot control system** for **path planning** and reduce oscillations between linear and adaptive motion processes.
- Tested different control strategies using ROS with C++ and python to evaluate mathematical models and existing framework.
- Enhanced robotic control workflows with testing new control settings via **RViz** and **Gazebo** simulations.

Multi-robot Collaborative Planning, Engineering Research Fellow June 2022 – August 2022
Berkeley Artificial Intelligence Research (BAIR) Lab, Advised by Prof. Claire J. Tomline *Berkeley, CA*

- Designed a motion planning process for a multi-robot research project, using Universal Robot 5 and Quadruped Robot to transport packages in a warehouse setting.
- Worked with Python in ROS, tested new paths for the UR5 using RViz and Gazebo simulations for real-time motion behavior and mapping different environments.
- Simulated motion stages for UR5 to use **visual perception** to find objects and perform grasp and pick-and-place to the quadruped robot.

Extended Opportunity Programs and Services, STEM Tutor January 2022 – May 2023
American River College *Sacramento, CA*

- Tutored college level students in Math, Physics, and Computer Sciences across diverse learning styles.
- contributed in creating and presenting informational workshops and tabling for new students, advocated for programs to helps student learning.
- **Gained leadership** and **communication skills** through working with diverse community of students from all backgrounds.

PROJECTS

Flexible Wearable Garment January 2025 – May 2025
Product Development Research Project *Berkeley, CA*

- Worked with a team of Ph.D. engineering students to develop a transformable textile that uses smock patterns to expand over 200% for programmable local and global adaptability to body geometry with tunable structural stability and breathability.
- We presented the design of the dress for pregnant women to potential users who have experienced pregnancy, aiming to collect their feedback on our design.
- Our product shows promise in applications across sportswear, children's clothing, and adjustable fashion accessories, thanks to its versatility and mechanical programmability.

Mechanical Engineering Capstone Project August 2024 – December 2024
Senior Capstone Project *Berkeley, CA*

- Worked with a team of engineering students to create a fully functional traditional tea making machine for our capstone project in Mechatronics course.
- Contributed in creating the CAD design in Fusion 360 for tea dispensing, testing electronics, PCB design, and programming the control system via C++.

Maze Navigator Robot January 2024 – May 2024
Personal Project *Berkeley, CA*

- Designed and implemented an IoT-based path navigation system using a rover robot to autonomously follow maze-like environments while monitoring surroundings.

- Optimized ultrasonic sensor response (Adafruit) for accurate boundary detection.
- Developed a wireless communication system (Wi-Fi) to transmit real-time sensor data, distance measurements, and path duration during navigation.

Biomedical Assistive Device for Parkinson's

September 2023 – February 2024

Mentored by Prof. Hannah Stuart

Berkeley, CA

- Co-designed a mechanical spoon using CAD to reduce hand tremor for people with Parkinson's and hand tremor.
- Tested polymer printing materials and 3D manufacturing; mechanical testing machines; 3D modeling in SolidWorks and Fusion.

A* Pathfinder Navigation in Different Environments

August 2023 – December 2023

Mentored by Dr. Jason J. Choi

Sacramento, CA

- Developed an A* (Astar) path finding algorithm for a Unitree Go1 robot to navigate in a city like simulated environment.
- Designed multiple simulated environments to test the algorithm, tested blocks and different path types to evaluate how fast the true path was found.
- Improved skills in robotic programming, algorithm design, testing and simulation skills; learned how to simulate Gazebo using ROS.

PUBLICATIONS

Evaluating Supernumerary Dorsal Grasping in the Home for People with C5–C7 Spinal Cord Injury

Andrew I.W. McPherson, Jungpyo Lee, **Alahe Akhavan**, Hannah S. Stuart.

IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) (Submitted for review)

Assistive Dorsal Grasper Modifications for In-Home Experiments

Alahe Akhavan, Andrew I.W. McPherson, Jungpyo Lee, Hannah S. Stuart.

Poster Presentation, SACNAS National Diversity in STEM Conference, 2024.

SERVICE/LEADERSHIP

Undergraduate Mentor

January 2024 – present

- Mentored incoming and transfer students to support their transition into UC Berkeley.
- Introduced mentees to university resources and encouraged participation in networking opportunities.
- Provided guidance on academic decisions while fostering confidence, motivation, and personal growth.

STEM Mentor and Volunteer

January 2025 – March 2025

- Mentored a group of diverse middle school students for their first scientific research experiment.
- Advocate for STEM through motivating research projects that are designed and completed by students.
- Helped young students gain communication, research, data collection, and analysis skills.

STEM Outreach

August 2024 – May 2025

- As a member of EDG Lab, partnered with disability organizations and managed information booths in the resource fairs to showcase past projects

- Organized a skill workshop on assistive devices for engineering students to gain manufacturing design, sensor testing, and circuit board design.

STEM Tutor

January 2022 – May 2023

- At my local community college, assisted people across a range of age groups and learning styles, helping them with STEM courses.
- Helped create workshops and regular check-ins to ensure and encourage student progress.

Volunteer Tutor

January 2017 – December 2021

- Thought and tutored low-income students of all ages with math and language at a Sunday school.

MENTORSHIP

Current Students

Jeremiah Avitia, Cristo Alberto Barragan, Gustavo Inga, Enkhjin Erdenee, September 2025 – present
Tessa Carol Brieger

Past Students

Cathy Sheng, Cassie Katherine Ly, Joya Mirabai Stewart, Alicia Wang January 2024 – Dec 2024

CONFERENCES

ASME IMECE2024 Conference Portland, Oregon

November 2024

- Poster Presentation: Poster Presentation: Assistive Dorsal Grasper Modifications for In-Home experiments

SACNAS Conference Arizona State University, Phoenix, Arizona

October 2024

- Poster Presentation: Assistive Dorsal Grasper Modifications for In-Home experiments

Undergraduate Women in Physics APS University of California, Santa Cruz

December 2022

Silicon Valley Women in Engineering Conference Virtual

March 2022

AWARDS

AUTODESK Instructables Award for Best Student Project (TremorEz)

September 2025

The Leadership Award TLA

May 2025

SACNAS National Diversity in STEM Conference Travel Scholarship

June 2024

UC Berkeley Mechanical Engineering Departmental Award

August 2023